

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL
UNIVERSITY, LONERE
WINTER SEMESTER EXAMINATION - Dec 2019**

B. Tech. Civil Engineering

Subject: Design of Steel Structures (BTCVC501)

Date: 09/12/2019

V Semester

Marks: 60

Time: 03 Hours

Instructions to the Students

1. Solve any five questions from Q. no. 1 to Q. no 6
2. Use of Non Programmable Scientific Calculator is allowed.
3. Assume Suitable data wherever necessary.
4. Use of IS 800:1984, IS 808, Steel table or IS handbook no. 1 and IS 875 is allowed.

Q.1 A) i) Write down the advantages and disadvantages of welded connections. **(06)**

ii) Explain various failure modes possible in a riveted joint with the suitable diagrams.

B) A joint in a truss girder consists of a double cover butt joint used to connect two bracing flats (ISF 200 x 12). The thickness of each cover plate is 8 mm. The flats have been joined by 9 power-driven field rivets in chain riveting pattern at a gauge of 50 mm as shown in the Figure 1. The nominal diameter of the rivets used is 22 mm. The yield strength of the flats used is 260 MPa. Find the capacity of the joint. **(06)**

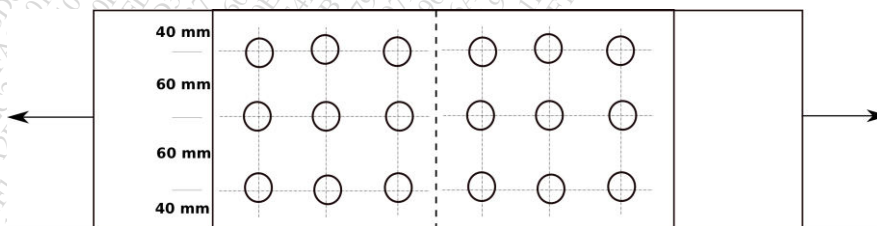


Figure 1

Q.2 A) A double angle tie ISA 150 x 115 x 10 mm ($f_y = 250$ MPa) with long leg connected to the same side of a gusset plate of 12 mm thickness by 18 mm diameter rivets, such that each angle is reduced in section by one rivets hole only. Determine the tensile strength of the member. **(06)**

B) A single angle discontinuous strut ISA 150 x 150 x 12 mm ($f_y = 250$ MPa) with single riveted connection is 3.5 m long. **(06)**

Determine the load carrying capacity of the compression member.

- Q.3** Design a beam of 5m effective span carrying a uniformly distributed load of 15kN/m, if the compression flange is laterally unsupported. ($f_y=250\text{MPa}$) assume dead load of the beam = 500 N/m. (12)

- Q.4** Solve any one question. (12)

- A)** Design a column with two channel sections (placed back-to-back) laced together to support a load of 750 kN. The effective length of the column is 10m. Provide single lacing system with riveted connection. Assume ($f_y=250\text{MPa}$).

OR

- B)** A column section I.S.H.B 350 @ 674 N/m carries an axial load of 1100kN. Design a suitable gusset base using riveted connection. (12)

- Q.5 A)** Determine the design forces on the panel points of a steel roof truss (as shown in Figure 2) 16 m in span and resting on brick masonry walls. The trusses are placed at 8 m c/c. The rise of truss is 1/4 of span. The roofing is of asbestos cement sheets of weight 171 N/m². The wind normal to roof is 940 N/m². (08)

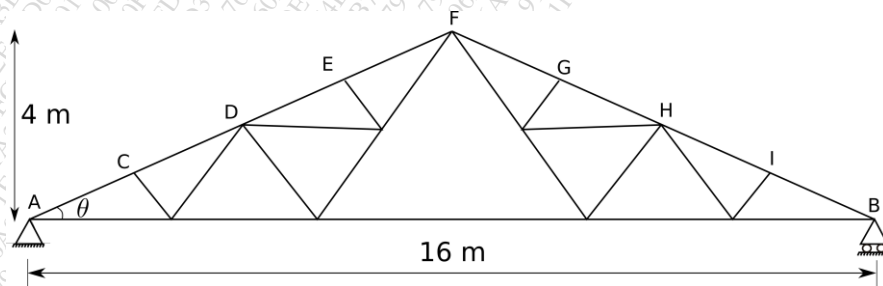


Figure 2

- B)** Explain in brief types of load acting on a gantry girder. (04)

- Q.6 A)** Define the plastic hinge. Find the length of plastic hinge for a simply supported rectangular beam subjected to a gradually increasing concentrated load P at the centre of its span. (08)

- B)** State the difference between working stress method and limit state method of design (04)

